

Analytically prove equality of asymptotic and worst-case IID k-unit prophet ratios

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Website #5 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Fix integers $k \geq 2$ and $n \geq 1$. For an IID nonnegative instance $I \in \mathcal{I}_{k,n}^{\text{IID}}$, let

$$\text{DP}(I)$$

be the value of the optimal adaptive online policy with capacity k , and let

$$\text{Proph}(I) = \mathbb{E} \left[\max_{S \subseteq [n], |S| \leq k} \sum_{i \in S} R_i \right]$$

be the prophet benchmark (the expected sum of the top k realized values).

The target quantity is

$$\gamma_{k,n} := \inf_{I \in \mathcal{I}_{k,n}^{\text{IID}}} \frac{\text{DP}(I)}{\text{Proph}(I)}.$$

The user asks for an analytic proof that, for every $k > 1$,

$$\lim_{n \rightarrow \infty} \gamma_{k,n} = \inf_{n \geq 1} \gamma_{k,n}.$$

I attack the strongest reading: the equality above for the fully adaptive online optimum DP against the prophet benchmark Proph.

Alternative readings considered

The statement is essentially unambiguous. The only nearby weaker readings are:

- replacing Proph by the ex-ante benchmark;
- replacing adaptive policies by static-threshold policies.

Those weaker variants are not the asked statement, but they are the strongest cases for which I obtained a complete proof.

Related literature found

- **Jiang–Ma–Zhang (2025)**, *Tightness Without Counterexamples* — PARTIAL for the exact problem, and TOOL for my proof. They characterize $\gamma_{k,n}$ for each fixed k, n via a semi-infinite LP, report numerical evidence that $\lim_{n \rightarrow \infty} \gamma_{k,n} = \inf_{n \geq 1} \gamma_{k,n}$, and explicitly state that an analytic proof of this equality for all $k > 1$ remained open in that paper. The same paper also proves that five other IID guarantees (including DP/ExAnte and all static-threshold variants) coincide and equal $\mathbb{E}[\min\{\text{Bin}(n, k/n), k\}]/k$. (columbia.edu)

- **Jiang–Ma–Zhang (2024), *Tight Guarantees for Multiunit Prophet Inequalities and Online Stochastic Knapsack*** — BACKGROUND. This is about the LP/ex-ante benchmark rather than the IID prophet benchmark, but it provides context for the multi-unit setting and the role of tight guarantees. (pubsonline.informs.org)
- **Brüstle–Perez-Salazar–Verdugo (2025), *Splitting Guarantees for Prophet Inequalities via Nonlinear Systems*** — PARTIAL. This gives analytical asymptotic lower bounds for the IID k -selection prophet inequality via a nonlinear system, i.e. genuine progress on $\lim_{n \rightarrow \infty} \gamma_{k,n}$, but not the finite-horizon equality with $\inf_n$. (pubsonline.informs.org)
- **Kong–Murthy (2026), *Multiunit I.I.D. Prophet Inequalities via Extreme Value Asymptotics*** — PARTIAL/BACKGROUND. This studies the asymptotic $n \rightarrow \infty$ regime through extreme-value methods, again relevant to the limit but not to the exact equality with $\inf_n$. (arxiv.org)

No consulted source claimed a full proof of

$$\lim_{n \rightarrow \infty} \gamma_{k,n} = \inf_{n \geq 1} \gamma_{k,n}$$

for the adaptive DP/Proph ratio and all $k > 1$. The strongest fully explicit formula available in the consulted literature is for the five weaker IID ratios mentioned above. (columbia.edu)

Proof sketch

I was not able to close the full adaptive DP/Proph statement. Instead, I prove the strongest complete analogue that *is* currently accessible from the exact fixed- (k, n) characterizations:

1. In the IID setting, the five quantities

$$\text{DP/ExAnte}, \quad \text{ST/Proph}, \quad \text{ST/ExAnte}, \quad \text{OST/Proph}, \quad \text{OST/ExAnte}$$

all have the same tight guarantee, namely

$$\sigma_{k,n} := \frac{\mathbb{E}[\min\{\text{Bin}(n, k/n), k\}]}{k} \quad (n > k),$$

by Jiang–Ma–Zhang. (columbia.edu)

2. So the only new work needed is to prove that $\sigma_{k,n}$ is nonincreasing in n .
3. I do this by an elementary “equalization lowers concave Poisson-binomial functionals” lemma. For

$$f(x) = \min\{x, k\},$$

which is concave on $\mathbb{Z}_{\geq 0}$, the functional

$$\Phi_m(p_1, \dots, p_m) := \mathbb{E} \left[f \left(\sum_{i=1}^m B_i \right) \right]$$

with $B_i \sim \text{Ber}(p_i)$ decreases when one moves a pair (p_i, p_j) toward equality while preserving their sum. Then I connect

$$\text{Bin}(n, k/n)$$

to the length- $(n+1)$ Poisson-binomial vector

$$(k/n, \dots, k/n, 0),$$

and explicitly transform this vector in n equalization steps into

$$(k/(n+1), \dots, k/(n+1)),$$

which has law $\text{Bin}(n+1, k/(n+1))$. Each step decreases Φ , hence $\sigma_{k,n} \geq \sigma_{k,n+1}$.

4. Therefore the worst horizon for those five ratios occurs at $n \rightarrow \infty$, and the common limit is the Poisson limit

$$\frac{\mathbb{E}[\min\{\text{Pois}(k), k\}]}{k}.$$

This does not settle the asked adaptive DP/Proph equality, but it does prove an exact horizon-monotonicity theorem for the strongest family of closely related IID guarantees that currently admits a closed form.

Main result

Theorem 1. *Fix $k \geq 2$. For $n \geq 1$, define*

$$\sigma_{k,n} := \inf_{I \in \mathcal{I}_{k,n}^{\text{IID}}} \frac{\text{ST}(I)}{\text{Proph}(I)},$$

where ST denotes the best (non-oblivious) static-threshold policy on the given IID instance.

Then the following hold.

1. If $n \leq k$, then

$$\sigma_{k,n} = 1.$$

2. If $n > k$, then the five IID tight guarantees

$$\inf_I \frac{\text{DP}(I)}{\text{ExAnte}(I)}, \quad \inf_I \frac{\text{ST}(I)}{\text{Proph}(I)}, \quad \inf_I \frac{\text{ST}(I)}{\text{ExAnte}(I)}, \quad \inf_I \frac{\text{OST}(I)}{\text{Proph}(I)}, \quad \inf_I \frac{\text{OST}(I)}{\text{ExAnte}(I)}$$

all coincide, and their common value is

$$\sigma_{k,n} = \frac{\mathbb{E}[\min\{\text{Bin}(n, k/n), k\}]}{k}.$$

3. The sequence $(\sigma_{k,n})_{n \geq 1}$ is nonincreasing. Hence

$$\lim_{n \rightarrow \infty} \sigma_{k,n} = \inf_{n \geq 1} \sigma_{k,n} = \frac{\mathbb{E}[\min\{\text{Pois}(k), k\}]}{k}.$$

4. Consequently, for the adaptive prophet-benchmark ratio,

$$\gamma_{k,n} \geq \sigma_{k,n} \quad \forall n \geq 1,$$

so in particular

$$\inf_{n \geq 1} \gamma_{k,n} \geq \frac{\mathbb{E}[\min\{\text{Pois}(k), k\}]}{k}.$$

Proof

Lemma 1. *Let $k \geq 1$, and define $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ by $f(x) = \min\{x, k\}$. Then f is concave in the discrete sense:*

$$f(t+2) - 2f(t+1) + f(t) \leq 0 \quad \forall t \geq 0.$$

For $m \geq 2$, write

$$\Phi_m(p_1, \dots, p_m) := \mathbb{E} \left[f \left(\sum_{i=1}^m B_i \right) \right],$$

where $B_i \sim \text{Ber}(p_i)$ are independent.

Fix indices $r \neq s$. If $a, b, a', b' \in [0, 1]$ satisfy

$$a + b = a' + b' \quad \text{and} \quad a'b' \geq ab,$$

then, with all other coordinates unchanged,

$$\Phi_m(\dots, a, \dots, b, \dots) \geq \Phi_m(\dots, a', \dots, b', \dots).$$

In words: for this concave functional, equalizing a pair of Bernoulli parameters weakly decreases the expectation.

Proof. The discrete concavity of $f(x) = \min\{x, k\}$ is immediate: the first differences are 1 up to $x = k - 1$ and 0 afterwards, so they are nonincreasing.

Now fix all Bernoulli variables except the r -th and s -th, and let

$$R := \sum_{i \notin \{r, s\}} B_i.$$

Conditioning on R , we obtain

$$\begin{aligned} & \mathbb{E}[f(R + B_r + B_s) \mid R] \\ &= (1-a)(1-b)f(R) + (a(1-b) + (1-a)b)f(R+1) + abf(R+2) \\ &= f(R) + (a+b)(f(R+1) - f(R)) + ab(f(R+2) - 2f(R+1) + f(R)). \end{aligned}$$

For fixed $a+b$, the only term depending on how the mass is split between a and b is the last one, and its coefficient

$$f(R+2) - 2f(R+1) + f(R)$$

is nonpositive by concavity. Therefore increasing the product ab weakly decreases the conditional expectation. Since $a'b' \geq ab$, we get

$$\mathbb{E}[f(R + B_r + B_s) \mid R] \geq \mathbb{E}[f(R + B'_r + B'_s) \mid R],$$

where $B'_r \sim \text{Ber}(a')$, $B'_s \sim \text{Ber}(b')$. Averaging over R yields the claim. $\square$

Proof. For $n \leq k$, every nonnegative arriving value can be accepted, so

$$\text{DP}(I) = \text{Proph}(I) = \text{ExAnte}(I)$$

for every IID instance I , and likewise the best static-threshold policies can accept everything. Hence every ratio in the theorem equals 1.

Assume now $n > k$. By Theorem 7, Theorem 8, and Corollary 2 of Jiang–Ma–Zhang, in the IID setting and for every fixed $k, n > k$, the five tight guarantees

$$\text{DP/ExAnte, ST/Proph, ST/ExAnte, OST/Proph, OST/ExAnte}$$

coincide, and their common value is exactly

$$\frac{\mathbb{E}[\min\{\text{Bin}(n, k/n), k\}]}{k}.$$

Hypothesis audit: those results are stated precisely for the IID setting, fixed k , and fixed horizon $n > k$, which is exactly our present regime. (columbia.edu)

Thus it remains only to prove that

$$a_{k,n} := \mathbb{E}[\min\{\text{Bin}(n, k/n), k\}]$$

is nonincreasing in n .

Set

$$\alpha_n := \frac{k}{n}, \quad \beta_n := \frac{k}{n+1}, \quad h_n := \alpha_n - \beta_n = \frac{k}{n(n+1)}.$$

Consider the length- $(n+1)$ vectors

$$p^{(0)} = (\underbrace{\alpha_n, \dots, \alpha_n}_{n \text{ times}}, 0),$$

and for $r = 1, \dots, n$,

$$p^{(r)} = (\underbrace{\beta_n, \dots, \beta_n}_{r \text{ times}}, \underbrace{\alpha_n, \dots, \alpha_n}_{n-r \text{ times}}, rh_n).$$

Then $p^{(n)} = (\beta_n, \dots, \beta_n)$.

I claim that for each $r = 1, \dots, n$,

$$\Phi_{n+1}(p^{(r-1)}) \geq \Phi_{n+1}(p^{(r)}),$$

where Φ_{n+1} is the functional from the lemma with $f(x) = \min\{x, k\}$.

Indeed, passing from $p^{(r-1)}$ to $p^{(r)}$ changes only one pair:

$$(\alpha_n, (r-1)h_n) \mapsto (\beta_n, rh_n).$$

These two pairs have the same sum because $\alpha_n = \beta_n + h_n$. Their products satisfy

$$\begin{aligned} \beta_n \cdot rh_n - \alpha_n \cdot (r-1)h_n &= h_n(r\beta_n - (r-1)\alpha_n) \\ &= h_n(r(\alpha_n - h_n) - (r-1)\alpha_n) \\ &= h_n(\alpha_n - rh_n). \end{aligned}$$

Now

$$rh_n = \frac{rk}{n(n+1)} \leq \frac{k}{n+1} < \frac{k}{n} = \alpha_n,$$

so $\alpha_n - rh_n > 0$, and therefore

$$\beta_n \cdot rh_n > \alpha_n \cdot (r-1)h_n.$$

By the lemma, equalizing this pair weakly decreases Φ_{n+1} , proving the claim.

Chaining the inequalities,

$$\Phi_{n+1}(p^{(0)}) \geq \Phi_{n+1}(p^{(1)}) \geq \dots \geq \Phi_{n+1}(p^{(n)}).$$

But $\Phi_{n+1}(p^{(0)})$ is exactly

$$\mathbb{E}[\min\{\text{Bin}(n, k/n), k\}],$$

because the last Bernoulli has parameter 0, while $\Phi_{n+1}(p^{(n)})$ is exactly

$$\mathbb{E}[\min\{\text{Bin}(n+1, k/(n+1)), k\}].$$

Hence

$$a_{k,n} \geq a_{k,n+1}.$$

Therefore $(a_{k,n})_{n>k}$ is nonincreasing, and so is

$$\sigma_{k,n} = \frac{a_{k,n}}{k}.$$

Since $0 \leq \min\{\text{Bin}(n, k/n), k\} \leq k$, dominated convergence together with the classical convergence

$$\text{Bin}(n, k/n) \Rightarrow \text{Pois}(k)$$

gives

$$\lim_{n \rightarrow \infty} \sigma_{k,n} = \frac{\mathbb{E}[\min\{\text{Pois}(k), k\}]}{k}.$$

A monotone sequence converges to its infimum, so

$$\lim_{n \rightarrow \infty} \sigma_{k,n} = \inf_{n \geq 1} \sigma_{k,n}.$$

Finally, for every instance I ,

$$\text{DP}(I) \geq \text{ST}(I),$$

hence

$$\gamma_{k,n} = \inf_I \frac{\text{DP}(I)}{\text{Proph}(I)} \geq \inf_I \frac{\text{ST}(I)}{\text{Proph}(I)} = \sigma_{k,n}.$$

Taking the infimum over n yields

$$\inf_{n \geq 1} \gamma_{k,n} \geq \inf_{n \geq 1} \sigma_{k,n} = \frac{\mathbb{E}[\min\{\text{Pois}(k), k\}]}{k}.$$

This proves the theorem. □

Gap to the full statement

The proved theorem does *not* establish

$$\lim_{n \rightarrow \infty} \gamma_{k,n} = \inf_{n \geq 1} \gamma_{k,n}$$

for the adaptive prophet-benchmark ratio $\gamma_{k,n}$.

Precisely what remains unproved is the horizon-monotonicity (or any substitute strong enough to imply worst-horizon-at-infinity) for the semi-infinite-LP value

$$\gamma_{k,n} = \inf_{I \in \mathcal{I}_{k,n}^{\text{IID}}} \frac{\text{DP}(I)}{\text{Proph}(I)}.$$

The consulted fixed- (k, n) characterization gives no explicit closed form for $\gamma_{k,n}$, only a semi-infinite LP plus a discretization-based numerical procedure; and Jiang–Ma–Zhang explicitly state that proving

$$\lim_{n \rightarrow \infty} \gamma_{k,n} = \inf_{n \geq 1} \gamma_{k,n}$$

analytically for all $k > 1$ remained open there. (columbia.edu)

A concrete route that still looks plausible to me is: prove either

- a direct contraction/compression map from the $n+1$ -horizon semi-infinite LP to the n -horizon one, or
- a block-splitting theorem that transfers any near-worst-case n -horizon IID instance to arbitrarily large horizons without improving the adaptive DP/Proph ratio.

My attempts at both ran into the same obstacle: when $k > 1$, a single block can contribute *multiple* useful values, so the $k = 1$ “max-root” argument no longer losslessly reduces a long horizon to a shorter one.

Self red-team

- **Counterexample attempt to the monotonicity lemma.** I checked the critical pairwise formula directly. For fixed R , the dependence on the chosen pair is exactly

$$ab[f(R+2) - 2f(R+1) + f(R)].$$

Because the bracket is ≤ 0 , increasing ab can only decrease the expectation. There is no hidden sign error.

- **Counterexample attempt to the horizon step.** The step

$$(\alpha_n, (r-1)h_n) \mapsto (\beta_n, rh_n)$$

preserves the pair sum and strictly increases the pair product, because

$$\beta_n rh_n - \alpha_n(r-1)h_n = h_n(\alpha_n - rh_n) > 0.$$

So every step goes in the correct direction.

- **Boundary cases.**

- If $n \leq k$, all ratios in the proved theorem are exactly 1, and I treated this separately.
- If $k = 1$, the same monotonicity proof for $\sigma_{1,n}$ still works, but the fully adaptive $\gamma_{1,n}$ statement is the classical Hill–Kertz/Correa story rather than the multi-unit question asked here.

- **Referee objection 1: “You used Jiang–Ma–Zhang only for $n > k$.”** Correct, and the proof explicitly isolates $n \leq k$ as a trivial case.
- **Referee objection 2: “Your theorem is weaker than the asked one.”** Correct; this is why the verdict is PARTIAL. I have kept the exact gap explicit rather than blurring it.
- **Referee objection 3: “Did you accidentally prove something false for adaptive DP/Proph?”** No. The monotonicity argument applies only after replacing $\gamma_{k,n}$ by the explicit closed form available for the five weaker IID ratios. I do *not* claim the same explicit formula for $\gamma_{k,n}$.

Ideas tried

1. **Direct monotonicity of $\gamma_{k,n}$ by deleting one period.** Idea: compress an $(n+1)$ -period adaptive policy into an n -period policy and compare ratios. Obstruction: for $k > 1$, deleting or merging periods does not preserve the possibility of taking several good items from a local cluster, while the prophet benchmark changes in a different way. I could not make the ratio comparison survive this mismatch.
2. **Block-max “root distribution” reduction.** Idea: given a hard n -horizon law F , take a law G with cdf $G = F^{1/m}$, so that the maximum of each block of m iid G -draws has law F . This exactly explains the $k = 1$ large-horizon reduction. Obstruction: for $k > 1$, a single block may contain several valuable items, so the long-horizon adaptive problem is not dominated by the block-maximum problem.
3. **Finite- n versus continuous-time/Poisson limit.** Idea: compare the fixed- n semi-infinite LP for $\gamma_{k,n}$ against its $n \rightarrow \infty$ Poissonized limit, hoping to prove a uniform lower bound

$$\gamma_{k,n} \geq \lim_{m \rightarrow \infty} \gamma_{k,m}.$$

Obstruction: the available asymptotic tools I found provide asymptotic lower bounds for large n , but I could not prove a clean domination valid for *every* finite n .

4. **What succeeded.** The exact Jiang–Ma–Zhang formula for the five weaker IID guarantees reduced the horizon question there to a pure binomial monotonicity statement, and that monotonicity can be proved by an elementary pairwise-equalization argument.

Invoked results

1. **Jiang–Ma–Zhang, Theorem 7 (IID reformulation).** T00L. For fixed k and $n > k$, the adversary’s optimization problems over IID type distributions are exactly reformulated as the corresponding semi-infinite IID LPs $\text{iidLPDP}/\text{ExAnte}_{k,n}$, $\text{iidLPOST}_{k,n}$, and $\text{iidLPST}_{k,n}$. (columbia.edu)

Hypothesis audit. I use it only in the IID setting, with fixed k and $n > k$, exactly as stated.

2. **Jiang–Ma–Zhang, Theorem 8.** T00L. For any fixed k and $n > k$,

$$\text{iidLPDP}/\text{ExAnte}_{k,n} = \sup_{\tau} \text{iidLPOST}(\tau)/\text{Proph}_{k,n} = \sup_{\tau} \text{iidLPOST}(\tau)/\text{ExAnte}_{k,n} = \frac{\mathbb{E}[\min\{\text{Bin}(n, k/n), k\}]}{k}$$

(columbia.edu)

Hypothesis audit. Again, I use it only for fixed k and $n > k$ in the IID setting, exactly matching the theorem.

3. **Jiang–Ma–Zhang, Corollary 2.** T00L. In the IID setting,

$$\text{iidLPST}/\text{Proph}_{k,n} = \sup_{\tau} \text{iidLPOST}(\tau)/\text{Proph}_{k,n}, \quad \text{iidLPST}/\text{ExAnte}_{k,n} = \sup_{\tau} \text{iidLPOST}(\tau)/\text{ExAnte}_{k,n}.$$

(columbia.edu)

Hypothesis audit. Used only in the same fixed- (k, n) , IID regime.

Self-confidence

3 — The partial theorem proved above is fully checked line by line; my uncertainty is only about the unproved extension from these five explicit IID guarantees to the full adaptive DP/Proph ratio.

Bibliography

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\begin{verbatim}
@article{JiangMaZhang2025TWC,
  author = {Jiashuo Jiang and Will Ma and Jiawei Zhang},
  title = {Tightness Without Counterexamples: A New Approach and New Results for Prophet Inequalities},
  journal = {Mathematics of Operations Research},
  year = {2025},
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  number = {2},
  pages = {956--987},
  doi = {10.1287/moor.2023.0221}
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@article{JiangMaZhang2024Multiunit,
  author = {Jiashuo Jiang and Will Ma and Jiawei Zhang},
  title = {Tight Guarantees for Multiunit Prophet Inequalities and Online Stochastic Knapsack},
  journal = {Operations Research},
  year = {2024},
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}

@article{BrustlePerezSalazarVerdugo2025,
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  journal = {Mathematics of Operations Research},
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@misc{KongMurthy2026,
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  title = {Multiunit I.I.D. Prophet Inequalities via Extreme Value Asymptotics},
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@article{HillKertz1982,
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  title  = {Comparisons of Stop Rule and Supremum Expectations of I.I.D. Random Variables},
  journal = {The Annals of Probability},
  year   = {1982},
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  doi    = {10.1214/aop/1176993886}
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